STAFF REVIEW · H446 115 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. An assembler.

2. It discards half the search space using an ordering comparison.

3. 1.

4. X is removed; Y remains at the front.

I Do — source worked steps

1. Model answer — Lexer identifies identifier, assignment, identifier, plus, integer. Parser checks grammatical structure. The compiler creates suitable target instructions; optimisation might fold constants only where semantics permit. — Check: link each claim to the scenario and use precise OCR terminology.

We Do — guided CHECK

1. What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.
Work through the method
Use the middle stage of the visual to follow the process and state each intermediate result.
Shared worked answer
Lexical analysis can still recognise tokens. Syntax analysis detects a missing closing delimiter because token order violates grammar. No valid final executable follows until corrected.

You Do — source mark schemes

1. Two marks each: tokenizer recognises lexemes/tokens; parser validates grammar/structure; generator outputs machine/target code.

2. Identifier score, assignment =, identifier score, operator +, integer literal 1.

3. Identify potential redundant addition [2], valid only if semantics identical [2], exceptions/type/overflow/side effects or floating nuances [2], reasoned qualified evaluation [2].

Exit Ticket — source answer

1. False. That is primarily lexical analysis.